

Business Insights Report

Capstone Sales & Customer Analytics Report

Executive Summary

The analysis of the customer and order datasets identified and Data quality issues such as missing values, replace null values and duplicate records were cleaned before analysis, ensuring reliable business insights.

Key Business Insights

1. Electronics is the Highest Revenue Category

- Electronics generated the highest total sales and profit among all product categories.
- It is the primary revenue driver for the business and contributes significantly to overall profitability.
- Electronics should remain the company's strategic focus for growth.

2. Sales and Profit Have a Positive Relationship

- Correlation analysis shows a **moderate positive correlation (≈ 0.61)** between Sales and Profit.
- Higher sales generally result in higher profits.
- Increasing sales in profitable categories is likely to improve overall business performance.

3. Several Orders are Loss-Making

- Analysis using NumPy identified multiple orders with **negative profit**.
- Many high-value sales still resulted in financial losses.
- High sales do not always guarantee profitability.

4. Missing Values and Duplicate Records Were Found

Data cleaning identified:

- Missing customer Region and Segment values.
- Missing Sales and Profit values.

- Duplicate Customer and Order records. These were corrected before analysis.
- Clean data improves reporting accuracy

5. Customer Sales are Unevenly Distributed

- The CTE and Window Function analysis showed that a small number of customers contribute a large portion of total sales.
- High-value customers generate significantly more revenue than average customers.

6. Sales Outliers Exist

The IQR method detected several unusually large sales transactions.

Actionable Recommendations

1. Increase investment in Electronics

- Expand inventory.
- Launch promotional campaigns.
- Introduce premium product bundles.

2. Investigate Loss-Making Orders

- Review pricing strategy.
- Reduce excessive discounts.
- Improve supplier negotiations.

3. Improve Data Quality

- Prevent duplicate Order IDs.
- Make Region and Segment mandatory fields.

4. Reward High-Value Customers

- Offer personalized discounts.
- Provide exclusive membership benefits.

5. Monitor Business Performance Using Power BI

Track KPIs such as:

- Total Sales
- Total Profit
- Profit Margin

- Order Count
- Average Order Value
- Sales by Category
- Sales by Region

6.Review Underperforming Categories

- Identify categories with low sales or low profit.
- Optimize pricing and marketing strategies.
- Remove consistently poor-performing products if necessary.